Convolutional Dictionary Updates as One Majorized Consensus Iteration: Exact Unification, Derived Equivalences, and Channel-Aware Splittings

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Abstract

Convolutional dictionary learning alternates sparse coding with a dictionary update whose frequency-domain normal equations are diagonal plus rank K. ADMM with iterated Sherman–Morrison or conjugate-gradient solves, ADMM consensus, the "3D" update and FISTA were derived separately for this structure. We show that they are settings of one majorized consensus iteration on a lifted problem: the observations are transformed by a unitary basis and partitioned into blocks, each block advances a copy of the dictionary by minimising a quadratic majorizer of its data term, and the copies are coupled by a multiplier or by substitution. A single implementation reproduces SPORCO's ISM, consensus and FISTA updates to 1.3×10^{-14} and its CG update to 6.8×10^{-6} in relative functional value at every iteration. Equivalences become identities between axis settings: the 3D update is consensus in the image-DFT basis (agreement 4.9×10^{-16} with SPORCO), colour channels treated as a third dimension give consensus in the channel-DFT basis, and the tying map that encodes a multi-channel representation decides whether channels raise the rank of the dictionary update or of sparse coding. The design space contains channel-aware splittings absent from existing implementations; they collapse exactly onto SPORCO at their degenerate settings, and on the dictionary-update subproblem the channel-blocked splittings converge $2.5\times$ faster than consensus on real colour data — because the filter-support constraint makes consensus reward homogeneous blocks, not because correlated channels should share one. Inside CDL these gains wash out: across 16 multi-channel regimes the updates need similar iteration counts, so the cost per iteration selects the update, while the channel correlation selects the representation (the shared dictionary fits independent channels better by up to 24.1%, the multi-channel dictionary fits RGB better by 19.9–24.0%).

1 Introduction

Convolutional sparse representations approximate a signal by a sum of filters convolved with sparse coefficient maps, $\mathbf{s} \approx \sum_m \mathbf{d}_m * \mathbf{x}_m$ [17, 35]. Learning the filters — convolutional dictionary learning (CDL) — alternates sparse coding with a dictionary update. Sparse coding is cheap because, in the DFT domain, its normal equations are diagonal plus rank one and yield to the Sherman–Morrison formula [29]. The dictionary update is the hard step: with K training images its systems are diagonal plus rank K, and a family of differently derived algorithms handles that structure — equality-constrained ADMM with iterated Sherman–Morrison (ISM) or conjugate-gradient (CG) inner solvers [31], spatial tiling, ADMM consensus and its "3D" frequency-domain variant [27], and FISTA [12], with further variants based on majorizers [6], accelerated consensus [25] and data partitioning [5]. In their comparative review, Garcia-Cardona and Wohlberg observed that two of

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these, derived independently, belong to one class, and that the differences between all of them are explained by the linear system each one solves [12]. Multi-channel signals add a second dimension to the problem: a colour image can be represented with one dictionary shared by all channels and per-channel maps, or with multi-channel filters and one map shared by the channels, and it is not obvious which update suits which representation.

This paper turns the observation about shared structure into a construction. We show that every batch dictionary update in the reference implementation, SPORCO [33], is one setting of a single iteration — a *majorized consensus iteration* on a lifted problem in which the training observations are transformed by a unitary basis, partitioned into blocks that each carry a copy of the dictionary, advanced by minimising a quadratic majorizer of their data term, and coupled either through a multiplier or by substitution (Sec. 3). The six axes of this iteration — coupling, partition, basis, local majorizer, inner solver, and the *tying map* through which each observation reads the dictionary — generate the known updates, expose their equivalences as identities between axis settings, and contain updates that nobody has implemented.

Contributions.

- **One iteration, one implementation.** A single driver instantiates ISM, CG, consensus and FISTA by axis values. Against SPORCO runs on identical MIRFLICKR inputs, the exact updates agree to 1.3×10^{-14} in relative functional value at every iteration and the CG update to 6.8×10^{-6} ($28/28$ runs within the 10^{-3} tolerance; Sec. 6.1).
- **Equivalences as axis algebra.** The 3D update is consensus in the image-DFT basis with penalty σ/K (agreement 4.9×10^{-16} with SPORCO run in three-dimensional mode, while plain consensus differs by 3.8×10^{-2}); treating colour channels as a third dimension is consensus in the channel-DFT basis (new; agreement 8.7×10^{-16}); bases acting inside blocks and partitions under substitution are inert; and the multi-channel-dictionary representation is equivariant under orthonormal colour transforms only with filter-joint normalisation (Sec. 4).
- **Where the channel index lives.** The tying map places the channel index either among the observations (shared dictionary: channels raise the rank of the dictionary update from K to CK) or among the variables (multi-channel dictionary: channels raise the rank of sparse coding to C), which makes the $K \leftrightarrow C$ duality of [12] an exact statement about the same linear systems. Both representations run from the same driver at the reference colour scale and reproduce SPORCO (Sec. 6.3).
- **Channel-aware splittings.** The grid contains channel-coupled consensus (one copy per image, exact rank-C coupling of its channels), channel-blocked and channel-basis splittings. We implement them and validate each by degenerate collapse onto SPORCO runs ($9/9$ checks within 9.4×10^{-16}). On the dictionary-update subproblem the channel-blocked splittings converge about $2.5\times$ faster than consensus on real colour data (51 iterations to a relative gap of 10^{-4}, against 153 for consensus and 31 for the far more expensive joint ISM update), while grouping an image's own channels (CCC) ties with grouping channels of different images when channels are independent and is slower when they are correlated. Inside CDL, with one inner iteration per stage, all updates with stable penalty rules end within 1.0% of each other (Secs. 5, 6.4).
- **A mechanism.** A frequency-decoupled model predicts that redundant (correlated) observations should share a block; the filter-support constraint couples the frequencies and reverses this: consensus is slow when blocks disagree, so every block should hold a representative sample of the data (Sec. 7).
- **A selection map** over the number of channels, the number of images and the inter-channel correlation, with a cost model derived from block ranks (Secs. 6.5, 9). The channel correlation decides the representation (the shared dictionary fits independent channels better by 6.9–24.1%, the multi-channel dictionary fits correlated and RGB channels better by up to 39.5%); inside CDL the updates need similar iteration counts, so the cost per iteration decides the update, and FISTA reaches a 1% tolerance first in 14 of 16 multi-channel regimes of the shared-dictionary representation.

2 Problem setting

Convolutional dictionary learning. Given training signals $\mathbf{s}_k \in \mathbb{R}^N$, $k = 1, \ldots, K$, CDL solves

$$\min_{\{\mathbf{d}_m\},\{\mathbf{x}_{m,k}\}} \frac{1}{2} \sum_k \Big\| \sum_m \mathbf{d}_m * \mathbf{x}_{m,k} - \mathbf{s}_k \Big\|_2^2 + \lambda \sum_{m,k} \|\mathbf{x}_{m,k}\|_1 \quad \text{s.t.} \quad \mathbf{d}_m \in C_{\mathrm{PN}} \ \ \forall m, \tag{1}$$

where C_{PN} contains the zero-padded filters that vanish outside an 8×8 support and have unit ℓ_2 norm; its projection $\Pi_{\mathcal{C}}$ is a support restriction followed by normalisation [12]. We hold fixed the functional, $\lambda = 0.1$, the constraint, the sparse-coding stage (ADMM CBPDN), and the coupling of the two stages (auxiliary variables are exchanged after a single iteration of each subproblem [11]). Everything we vary lives inside the dictionary update.

The dictionary update over an observation index. To treat greyscale and multi-channel data in one notation, we index the data by a finite set J of *observations*. Observation $j \in J$ has a signal $\mathbf{s}_j \in \mathbb{R}^N$, coefficient maps $\{\mathbf{x}_{j,m}\}_{m=1}^M$ supplied by the sparse-coding stage, and reads the dictionary parameter Θ through a linear *tying map* T_j. The dictionary update is

$$\min_{\Theta} \ f(\Theta) + \iota_{\mathcal{C}}(\Theta), \qquad f(\Theta) = \frac{1}{2} \sum_{j \in J} \Big\| \sum_m \mathbf{x}_{j,m} * (T_j \Theta)_m - \mathbf{s}_j \Big\|_2^2. \tag{2}$$

For greyscale data $J = \{1, \ldots, K\}$ and $T_j = I$. For C-channel signals there are two representations [30], which differ only in their tying map:

- **R1 (shared single-channel dictionary, per-channel maps).** $J = \{1..C\} \times \{1..K\}$, $\Theta = \{\mathbf{d}_m\}$, $T_{(c,k)} = I$, $\mathbf{x}_{(c,k),m} = \mathbf{x}_{c,k,m}$: each channel of each image is one observation.
- **R2 (multi-channel dictionary, shared maps).** $J = \{1..C\} \times \{1..K\}$, $\Theta = \{\mathbf{d}_{c,m}\}$, $T_{(c,k)}\Theta = \{\mathbf{d}_{c,m}\}_m$, $\mathbf{x}_{(c,k),m} = \mathbf{x}_{k,m}$: channel c of image k sees the channel-c slice of every filter, and all channels of an image share one map.

A third representation couples the per-channel maps of R1 through an $\ell_{2,1}$ penalty across channels (R3, joint sparsity [30]); its dictionary update is that of R1. For R2 we use, as the reference implementation does, the normalisation $\|\mathbf{d}_m\|_2 = 1$ jointly over the channels of filter m; the per-channel variant $\|\mathbf{d}_{c,m}\|_2 = 1$ written in the reviewed paper's Eq. (80) is a different constraint set (Sec. 4).

Frequency-domain structure. Let $\hat{\cdot}$ denote the 2-D DFT and $\mathbf{a}_j(n) = (\hat{x}_{j,1}(n), \ldots, \hat{x}_{j,M}(n))^\top \in \mathbb{C}^M$ the row of observation j at frequency n. By Parseval, f decouples over frequencies into $\frac{1}{2N} \sum_n \sum_j |\mathbf{a}_j(n)^\top \widehat{T_j \Theta}(n) - \hat{s}_j(n)|^2$, so the curvature of any group $B \subseteq J$ of observations with a common tying map is, at every frequency,

$$Q_B(n) = \sum_{j \in B} \overline{\mathbf{a}_j(n)}\, \mathbf{a}_j(n)^\top, \qquad \operatorname{rank} Q_B(n) \le |B|. \tag{3}$$

For greyscale data and $B = J$ this is the familiar "diagonal plus rank K" structure that precludes a single Sherman–Morrison solve. Every dictionary update in use is a way of splitting, approximating, or avoiding systems $(\sigma I + Q_B(n))\boldsymbol{\theta} = \mathbf{b}$.

3 One iteration behind all dictionary updates

3.1 Lifting: a basis, a partition, and copies of the dictionary

Let U be a unitary matrix acting on the observation index (mixing only observations that share a tying map), and define transformed rows and signals $\tilde{\mathbf{a}}_\ell = \sum_j U_{\ell j} \mathbf{a}_j$, $\tilde{s}_\ell = \sum_j U_{\ell j} \hat{s}_j$.

Lemma 1 (basis invariance)**.** *For every $\boldsymbol{\theta}$ and frequency n, $\sum_\ell |\tilde{\mathbf{a}}_\ell^\top \boldsymbol{\theta} - \tilde{s}_\ell|^2 = \sum_j |\mathbf{a}_j^\top \boldsymbol{\theta} - \hat{s}_j|^2$.*

The residual vector is multiplied by U, which preserves its norm. Now partition the transformed observations into blocks $\mathcal{P} = \{B_1, \ldots, B_P\}$, write f_p for the data term of block p, and give each block its own copy Θ_p of the dictionary, tied to a consensus variable G:

$$\min_{\Theta_1, \ldots, \Theta_P, G} \ \sum_{p=1}^P f_p(\Theta_p) + \iota_{\mathcal{C}}(G) \quad \text{s.t.} \quad \Theta_p = G, \ \ p = 1, \ldots, P. \tag{4}$$

Table 1: Design axes of the majorized consensus iteration.

Axis	Symbol	Values used in this paper
Coupling enforcement	β	1: multiplier (penalty σ, relaxation α, optional residual balancing); 0: substitution (no multiplier; step $1/L$, Nesterov schedule ϑ_i)
Partition of observations	$\mathcal{P}$	joint ($P=1$), image (all channels of one image), channel (one channel of all images), single (one observation), groups of $g_c \times g_k$
Observation basis	U	identity; DFT over images; DFT over channels; KLT over channels
Local model	M_p	exact $\nabla^2 f_p$; isotropic $(L/P)\,I$
Inner solver (exact model)	$\mathcal{S}$	iterated Sherman–Morrison recursion (rank r; $r=1$ is the Sherman–Morrison formula); conjugate gradient (tolerance, warm start)
Representation	T_j	greyscale, R1 (shared dictionary), R2 (channel dictionaries); normalisation group: filter (joint over channels) or channel

By Lemma 1, $\sum_p f_p = f$, so (4) is equivalent to (2) for every choice of $(U, \mathcal{P})$.

3.2 The majorized consensus iteration

Each block is advanced by minimising a quadratic model of f_p around an expansion point E_p, $q_p(\Theta; E_p) = f_p(E_p) + \langle \nabla f_p(E_p), \Theta - E_p \rangle + \frac{1}{2}\|\Theta - E_p\|^2_{M_p}$ with $M_p \succeq \nabla^2 f_p$ (a majorizer), and the copies are coupled through a penalty $\sigma \geq 0$ and, optionally, a scaled multiplier H_p:

$$\Theta_p^+ = \arg\min_{\Theta}\ q_p(\Theta; E_p) + \tfrac{\sigma}{2}\|\Theta - G + H_p\|^2, \tag{5a}$$

$$R_p = \alpha\,\Theta_p^+ + (1-\alpha)\,G, \tag{5b}$$

$$G^+ = \Pi_{\mathcal{C}}\Big(\tfrac{1}{P}\textstyle\sum_p \big(R_p + \beta H_p\big)\Big), \tag{5c}$$

$$H_p^+ = \beta\,\big(H_p + R_p - G^+\big), \tag{5d}$$

$$E_p^+ = G^+ + \vartheta_i\,(G^+ - G). \tag{5e}$$

We call (5) the *majorized consensus iteration* (MCI). One outer CDL iteration performs exactly one MCI step. With the exact model $M_p = \nabla^2 f_p$ the local step (5a) no longer depends on E_p and becomes the linear system

$$\big(\sigma I + Q_{B_p}(n)\big)\,\hat{\boldsymbol{\theta}}_p(n) = \textstyle\sum_{\ell \in B_p} \overline{\tilde{\mathbf{a}}_\ell(n)}\,\tilde{s}_\ell(n) + \sigma\big(\hat{G}(n) - \hat{H}_p(n)\big) \quad \text{at every frequency } n, \tag{6}$$

a diagonal-plus-rank-$|B_p|$ system. With the isotropic model $M_p = (L/P)\,I$ it is the closed form $\Theta_p^+ = \big((L/P)E_p - \nabla f_p(E_p) + \sigma(G - H_p)\big)/(L/P + \sigma)$.

3.3 The known updates are settings of the axes

Proposition 1 (recovered updates)**.** *With α and the penalty schedule as in the reference implementation:*

1. *β=1, $\mathcal{P}$=joint, U=I, exact model: ADMM with the equality constraint $\mathbf{d} = \mathbf{g}$; the inner solver is ISM or CG.*
2. *β=1, $\mathcal{P}$=single (greyscale: one block per image), U=I, exact model: ADMM consensus; every local system has rank one and the recursion reduces to Sherman–Morrison.*
3. *β=0, σ=0, isotropic model, Nesterov $\vartheta_i = (t_i - 1)/t_{i+1}$: FISTA, $G^+ = \Pi_{\mathcal{C}}(E - \nabla f(E)/L)$, for every $(\mathcal{P}, U)$.*
4. *β=1, $\mathcal{P}$=single, U=DFT over images, exact model: the "3D" update of Šorel and Šroubek [27] (Sec. 4).*

Items 1–2 follow by substituting the axis values into (5): with one block, (5c)–(5d) are the $\mathbf{g}$- and $\mathbf{h}$-steps of equality-constrained ADMM; with one block per image they are the averaging-plus-projection and dual steps of consensus ADMM. For item 3, with $\beta = 0$ there is no multiplier, $\sigma = 0$ and the isotropic model give $\Theta_p^+ = E - (P/L)\nabla f_p(E)$, and the average in (5c) is $E - \nabla f(E)/L$ because $\sum_p \nabla f_p = \nabla f$ (Lemma 1). The multiplier initialisation is the only convention that differs

between the reference classes ($H_0 = G_0$ for ISM/CG, $H_0 = G_0/\sigma$ for consensus); both lie in the normal cone of the unit sphere at G_0, and we expose it as $H_0 = \kappa G_0$.

3.4 Which cells are consistent

Proposition 2 (fixed points). *(i) If $\beta = 1$ (either local model) and $\alpha \neq 0$, every fixed point has $\Theta_p = G$ and $-\nabla f(G) \in N_{\mathcal{C}}(G)$, i.e. G is a first-order stationary point of (2). (ii) If $\beta = 0$ with the isotropic model and $\sigma = 0$, fixed points satisfy $G = \Pi_{\mathcal{C}}(G - \nabla f(G)/L)$ (stationary). (iii) If $\beta = 0$ with the exact model and $\sigma > 0$, fixed points satisfy $G = \Pi_{\mathcal{C}}\big(\frac{1}{P}\sum_p \operatorname{prox}_{f_p/\sigma}(G)\big)$: stationary points of a Moreau-smoothed objective, not of (2). (iv) With the exact model and $\sigma = 0$ the local systems are singular whenever $|B_p| < M$.*

For (i): at a fixed point $H_p^+ = H_p$ forces $R_p = G^+ = G$, hence $\Theta_p = G$; the local optimality condition of (5a) at $E_p = \Theta_p = G$ gives $\nabla f_p(G) = -\sigma H_p$ for both models, and (5c) gives $\frac{1}{P}\sum_p H_p \in N_{\mathcal{C}}(G)$; summing yields the claim. (iii) is backward–backward splitting. The design space therefore contains two consistent "regimes" of the coupling axis — a multiplier with any local model, or substitution with an isotropic model — and one inconsistent corner (exact model without multiplier), which we run as a probe in Sec. 8.

4 Equivalences that follow from the axis algebra

Because every update is the same iteration, two updates coincide exactly when their axis settings induce the same local systems (6) and the same averaging. Several consequences are immediate: two generic identities (Proposition 3), the 3D update as a change of basis, and statements about channels (Propositions 4 and 5).

Proposition 3 (inert axes). *(a) If the basis acts inside blocks, $U = \bigoplus_p U_p$, the iterates coincide with those for $U = I$: $Q_{\tilde{B}_p} = \tilde{A}_p^H \tilde{A}_p = A_p^H U_p^H U_p A_p = Q_{B_p}$ and the right-hand sides agree. (b) With $\beta = 0$, $\sigma = 0$ and the isotropic model, the iterates do not depend on $(\mathcal{P}, U)$ (proof of Proposition 1, item 3).*

The "3D" update is consensus in the image-DFT basis. The 3D update stacks the K images into one volume, learns three-dimensional filters constrained to the first slice, and runs equality-constrained ADMM on the volume, where each three-dimensional frequency gives a rank-one system [27]. Apply the unnormalised DFT F_K along the image axis: the volume's data term becomes $\frac{1}{K}\sum_\omega \frac{1}{2}\|X''_\omega \mathbf{d}''_\omega - \mathbf{s}''_\omega\|^2$ with $X''_\omega = \sqrt{K}\,\tilde{X}_\omega$ the convolution with the ω-th unitary-DFT combination of the maps, the slice-0 constraint on $\mathbf{g}$ becomes $\mathbf{d}''_\omega = \mathbf{g}$ for every ω, and the penalty $\frac{\sigma}{2}\|\mathbf{d} - \mathbf{g} + \mathbf{h}\|^2$ becomes $\frac{\sigma}{2K}\sum_\omega \|\mathbf{d}''_\omega - \mathbf{g} + \mathbf{h}''_\omega\|^2$. Dividing by the common factor, the 3D update is the MCI with $\beta = 1$, $U =$ unitary DFT over images, $\mathcal{P} =$ single, exact model and penalty σ/K — the reviewed paper's observation that 3D is "consensus in the DFT domain" [12], here obtained as a change of a single axis value and made exact, including the penalty map. It also shows that 3D and consensus are *different* cells (same partition, different basis), which is why the paper found them to behave "similarly" rather than identically.

Treating colour channels as a third dimension. The same construction applied to the *channel* axis of R1 is new. Since R1 reads one single-channel filter in every channel, its data term equals that of a three-dimensional convolutional model over (space $\times$ channel) whose filters are confined to channel slice 0. Running equality-constrained ADMM on that volume model (one dictionary, one copy per image or a single joint copy) and transforming along the channel axis gives:

Proposition 4 (channel-Fourier cells). *For R1, equality-constrained ADMM on the channel volume equals the MCI with $U =$ unitary DFT over channels and $\mathcal{P} =$ channel (one block per channel frequency, holding all K images; rank K); volume-wise consensus over images equals $U =$ DFT over channels with $\mathcal{P} =$ single. In both cases the penalty maps as $\sigma \mapsto \sigma/C$.*

Where the channel index lives. The tying map decides on which side of the alternation the channel index acts.

Proposition 5 (representations and the $K \leftrightarrow C$ duality)**.** *(a) R1's dictionary update is the greyscale update on the CK channel images (all $T_j = I$); with plain ℓ_1 sparse coding, which decouples over channels, the whole R1 CDL run coincides with greyscale CDL on CK images. Channels therefore raise the rank of the dictionary-update systems from K to CK. (b) In R2 the curvature is $I_C \otimes \left(\sum_k \overline{\mathbf{a}_k}\mathbf{a}_k^\top\right)$: one rank-$K$ factor shared by C right-hand sides, so channels do not raise the dictionary-update rank; instead the sparse-coding system at each frequency is $(\rho I + \sum_c \overline{\hat{\mathbf{d}}_c}\hat{\mathbf{d}}_c^\top)\hat{\mathbf{x}} = \mathbf{b}$, which has exactly the form (6) with $(\hat{\mathbf{d}}_c, C, \rho)$ in place of $(\mathbf{a}_k, K, \sigma)$. (c) With filter-joint normalisation, R2 is equivariant under orthonormal colour transforms $V \in O(C)$: running CDL on $(V\mathbf{s}, V\mathbf{d}^{(0)})$ produces $(V\mathbf{d}^{(i)}, \mathbf{x}^{(i)})$ and identical functional values. Per-channel normalisation, and the per-channel ℓ_1 term of R1, break this invariance (except for signed permutations).*

Part (b) is the $K \leftrightarrow C$ duality of the reviewed paper made exact: every inner solver, partition and basis of the template is also a sparse-coding solver for R2 (the reference implementation's multi-channel CBPDN already calls the ISM kernel over the channel axis). Part (c) follows because the CBPDN systems depend on D only through $\hat{D}^H\hat{D}$ and $\hat{D}^H\hat{S}$, which are invariant under V, and because $\Pi_{\mathcal{C}}$ commutes with V when the norm is taken jointly over channels. In the space×channel picture, R1 and R2 are the two slice restrictions of one circular convolutional model — R1 confines the *filters* to channel slice 0, R2 confines the *maps* — and the channel DFT turns R1's update into consensus over channel frequencies (Proposition 4) and R2's into C problems sharing one factor.

5 Channel-aware splittings predicted by the design space

For R1 the observation index is a product, channels × images, and the reference implementation offers only its two extremes: one joint block holding all CK observations (ISM, CG) or CK singleton blocks (consensus, where channels are simply appended to the image axis). The partition and basis axes fill in the rest of the grid (Table 2); the cells below are, to our knowledge, not implemented in existing software and not described in the literature for multi-channel CDL.

- **Channel-coupled consensus (CCC):** $\mathcal{P}$ = image. One dictionary copy per image; its local system couples that image's C channel observations exactly (rank C, iterated Sherman–Morrison). The local solve is the transpose of the multi-channel sparse-coding solve (Proposition 5b): the same rank-C recursion, with coefficient spectra in place of filter spectra. Like consensus it parallelises over images.
- **Channel-blocked joint update (ChB):** $\mathcal{P}$ = channel. One copy per colour channel, each solving a rank-K system; consensus acts only across the C channels.
- **Channel-Fourier cells (CF):** U = DFT over channels with $\mathcal{P}$ = single or channel. By Proposition 4 these are the channel-volume ("3D over channels") updates.
- **KLT cells:** U = the orthonormal eigenbasis of the channel second-moment matrix of the training signals (luminance/chrominance-like components), with $\mathcal{P}$ = single or channel.

Each new cell has degenerate settings that land on an existing update (Table 6): CCC with channel-group size one is per-observation consensus, CCC on greyscale data is consensus, CCC on a single image is ISM; ChB on greyscale data is ISM and on a single image is consensus; CF-single and KLT-single with $U = I$ are consensus. Grouping images, $\mathcal{P}$ = groups of g_k images, is the partition used by consensus convolutional sparse coding [5] and is not counted as new.

Cost of a cell. Per outer iteration and dictionary channel, the dictionary update costs $(C_x KM + 2M)$ FFTs of size N (coefficient spectra, and the round trip of the consensus variable), plus the local solves: with blocks of size r the Sherman–Morrison recursion costs $O(r^2MN)$ per block and $O(|J|\, r\, MN)$ in total, CG costs $O(n_{\mathrm{cg}}|J|MN)$, and the isotropic model $O(|J|MN)$. The partition therefore trades block rank r (cost linear in r per observation) against the number of copies $P = |J|/r$ (memory $O(PMN)$, and — as we show in Sec. 7 — convergence). For R1 the four named partitions have $r \in \{CK, K, C, 1\}$; for R2 the shared factor means that C right-hand sides cost one recursion.

Table 2: The design space for the shared-dictionary representation R1 (rows: partition of the $C \times K$ observations, with block rank r; columns: observation basis), with multiplier coupling ($\beta = 1$) and the exact local model. REC: recovered and validated against SPORCO; NEW: implemented here, validated by degenerate collapse; $\equiv$: identical iterates to the named cell (axis identity, verified numerically: E-labels refer to Table 4); CTL: control. Not shown: with $\beta = 0$ and the isotropic model every row and column is FISTA (REC, E3); $\beta = 1$ with the isotropic model is linearised consensus (cf. [25]); $\beta = 0$ with the exact model is inconsistent (Proposition 2iii). For greyscale data, (single, DFT over images) is the 3D update (E1) and groups of images are the CCSC partition [5].

Partition	r	identity	DFT over channels	KLT over channels	images permuted per channel
joint	CK	ISM, CG REC	$\equiv$ ISM (E3)	$\equiv$ ISM (E3)	$\equiv$ ISM (E3)
channel	K	ChB NEW	CF-channel NEW (E2)	KLT-channel NEW	$\equiv$ ChB (E3)
image	C	CCC NEW	$\equiv$ CCC (E3)	$\equiv$ CCC (E3)	CCC-shuffled CTL
single	1	consensus REC	CF-single NEW (E2)	KLT-single NEW	$\equiv$ consensus (E3)

6 Experiments

Data and protocol. All training data come from MIRFLICKR-25k [15], read from the original archive and preprocessed as in the reviewed paper: centre crop, rescale to 256×256 (and, for the small greyscale and controlled studies, further to 128×128), conversion to greyscale where required, division by 255, and high-pass filtering by subtracting a Tikhonov low-pass component with regularisation 5, using SPORCO's own filter. Images are drawn by a logged seed (colour experiments reject near-greyscale images). We use 8×8 filters, $\lambda = 0.1$, auxiliary-variable coupling with one iteration of each stage, and SPORCO's ADMM CBPDN for sparse coding (penalty $\rho = 2.2$ for joint and FISTA cells, 3.0 for consensus-type cells, the reviewed paper's guidelines). The functional is evaluated as in SPORCO with `AccurateDFid`: at the consensus (or projected) dictionary and the CBPDN auxiliary variable, with filters of unit norm. Template runs and references always receive identical inputs (images, initial dictionary, λ, penalties, inner-solver settings).

Implementation. The template is one class (`mci/template.py`, 390 lines including documentation, of which about 270 are code, plus 90 lines of inner solvers) that implements (5) with SPORCO's dictionary-update interface; it never calls a SPORCO dictionary-update solver. Each named update is a dictionary of axis values (`mci/cells.py`). It runs on NumPy (CPU) or CuPy (GPU). SPORCO references run on the CPU, and for the large colour runs on the GPU through SPORCO's own CuPy patching mechanism applied to its dictionary-update modules; GPU and CPU SPORCO agree to $1.4{\times}10^{-15}$ (functional) and $8.0{\times}10^{-14}$ (dictionary) on MIRFLICKR data, and the GPU template agrees with CPU SPORCO to $1.4{\times}10^{-15}$. Penalties for the new cells, which have no published rules, were chosen by the reviewed paper's protocol (lowest functional after a fixed number of iterations over a logarithmic grid) on real colour data at $K \in \{10, 20\}$, and scaling rules in K were derived from the two optima; the same search was run for the known cells.

6.1 The known updates as axis settings

For each of SPORCO's four dispatchable families (ISM, CG, consensus, FISTA) we ran SPORCO's `ConvBPDNDictLearn` and the corresponding template instance on the same greyscale inputs ($K = 5$ MIRFLICKR images of 128×128, $M = 32$, 200 iterations) for three seeds and two configurations: the reviewed paper's protocol (fixed penalties, no relaxation; $\sigma = 0.5K + 7$ for ISM/CG, $\sigma = 2.2$ for consensus, $L = 14K$ for FISTA) and SPORCO's library dynamics (the same penalties with SPORCO's defaults otherwise: over-relaxation 1.8 and residual balancing every 10 iterations where SPORCO enables it). The template reads those effective settings from SPORCO's option objects; the CG tolerance (10^{-3}), iteration cap (1000) and warm start are SPORCO's. One further run uses $K = 10$. All 28 runs pass the 10^{-3} criterion on both the trajectory and the final dictionary (Fig. 1, Table 3). The deviations of the ISM, consensus and FISTA instances are at the level of floating-point round-off ($\leq 1.3{\times}10^{-14}$ in the functional, $\leq 4.3{\times}10^{-13}$ in the dictionary), i.e. the template does not approximate these updates — it executes the same iterations. CG agrees to $6.8{\times}10^{-6}$ / $4.4{\times}10^{-4}$, for the reason given in Sec. 8. With a spatial mask (25% of the pixels, SPORCO's mask-decoupled sparse coding), the template's FISTA cell with the masked gradient matches SPORCO's masked dictionary learning to $8.0{\times}10^{-16}$ / $1.0{\times}10^{-13}$ over three seeds.

Table 3: Updates and representations as instances of the majorized consensus iteration. Deviation: maximum relative functional deviation over all logged iterations / relative Frobenius deviation of the final dictionary, against the reference named in the text. Greyscale rows: SPORCO `ConvBPDNDictLearn` (worst case over 3 seeds, 2 protocols, and the $K = 10$ run; $K = 5$, 128×128, $M = 32$); 3D: SPORCO run in three-dimensional mode. Colour rows ($K = 40$ MIRFLICKR colour images, 256×256, $M = 64$, 500 iterations, seed 0): SPORCO references at $\lambda = 0.1$ with identical penalties; (s): no reference at this scale, value from the small-scale GPU check; R3 reference: SPORCO `DictLearn` with `ConvBPDNJoint` and consensus. New cells: worst degenerate-collapse deviation ((d), Table 6) or deviation from SPORCO's three-dimensional mode ((e), Table 4). Final F: functional at the last iteration (200 for greyscale, 150 for 3D, 500 for colour; R3 includes $\mu\|\mathbf{x}\|_{2,1}$, $\mu = 0.1$).

Method	Repr.	Axis settings	Deviation vs. reference	Final F
ISM	grey	β=1, joint, I, exact, ISM (r=K)	1.4×10^{-15} / 1.2×10^{-13}	62.13
CG	grey	β=1, joint, I, exact, CG (τ=10^{-3})	6.8×10^{-6} / 4.4×10^{-4}	62.14
Consensus	grey	β=1, single, I, exact, SM (r=1)	9.4×10^{-16} / 7.1×10^{-14}	62.06
FISTA	grey	β=0, isotropic, Nesterov	1.3×10^{-14} / 4.3×10^{-13}	61.61
3D (SPORCO volume)	grey	β=1, single, $\mathrm{DFT}_{\mathrm{img}}$, exact, SM	4.9×10^{-16} / 1.4×10^{-14}	62.72
FISTA, masked data term	grey	β=0, isotropic, Nesterov, mask W in ∇f_p	8.0×10^{-16} / 1.0×10^{-13}	53.49
Consensus	R2	β=1, image, I, exact, SM, shared factor	1.3×10^{-15} / 1.2×10^{-13}	6406.6
FISTA	R2	β=0, isotropic, Nesterov	4.2×10^{-16} / 1.2×10^{-14}	6383.1
ISM	R2	β=1, joint, I, exact, ISM (r=K), shared factor	7.3×10^{-16} / 1.1×10^{-14} (s)	6380.9
Consensus	R1	β=1, single, I, exact, SM	4.5×10^{-16} / 3.4×10^{-14}	8101.8
FISTA	R1	β=0, isotropic, Nesterov	2.0×10^{-15} / 1.1×10^{-13}	8092.6
ISM	R1	β=1, joint, I, exact, ISM (r=CK)	1.4×10^{-15} / 1.6×10^{-14} (s)	8097.4
Consensus, $\ell_{2,1}$ coding	R3	as R1 consensus	1.6×10^{-15} / 4.8×10^{-13}	11531.0
CCC (new)	R1	β=1, image, I, exact, ISM (r=C)	9.4×10^{-16} (d)	8111.2
ChB (new)	R1	β=1, channel, I, exact, ISM (r=K)	6.1×10^{-16} (d)	8114.8
CF-single (new)	R1	β=1, single, $\mathrm{DFT}_{\mathrm{ch}}$, exact, SM	8.7×10^{-16} (d)	8132.1
CF-channel (new)	R1	β=1, channel, $\mathrm{DFT}_{\mathrm{ch}}$, exact, ISM	6.4×10^{-16} (e)	8115.2
KLT-single (new)	R1	β=1, single, $\mathrm{KLT}_{\mathrm{ch}}$, exact, SM	9.4×10^{-16} (d)	8170.8

6.2 Equivalences

Table 4 verifies each identity of Sec. 4 on MIRFLICKR data ($K = 5$, 128×128, $M = 32$, 150 iterations). The 3D update was run as SPORCO itself: the K greyscale images form one volume, the filters have support $8 \times 8 \times 1$, and SPORCO's equality-constrained update runs in three dimensions. Its trajectory coincides with the template cell (single blocks, image-DFT basis, penalty σ/K) to 4.9×10^{-16}, whereas plain consensus with the same penalty differs by 3.8×10^{-2}. The channel analogue — SPORCO in three-dimensional mode along the colour axis of R1 — coincides with the channel-Fourier cells to 8.7×10^{-16} for both its joint and its consensus update. All axis identities hold to 8.7×10^{-16} or better, and every contrast row, where the identity is *not* predicted, differs by at least 1.5×10^{-2}.

6.3 Both colour representations from one driver, at the reference scale

We learned $M = 64$ filters (8×8, and $8 \times 8 \times 3$ for R2) from $K = 40$ colour MIRFLICKR images of 256×256 pixels for 500 iterations at $\lambda = 0.1$, the scale of the reviewed paper's colour experiment, using the same template driver for both representations: R2 differs from R1 only in its tying map (Sec. 2). References are SPORCO's `ConvBPDNDictLearn` at $\lambda = 0.1$ on the multi-channel-dictionary path (R2) and on the single-channel-dictionary path with plain CBPDN (R1), with the same images, initial dictionary and penalties (the paper's rules, including its colour adjustment $L = 3 \times 14K$ for FISTA).

Agreement and convergence. Fig. 2 overlays the template instances on the references. For R2 the consensus and FISTA instances agree with SPORCO to 1.3×10^{-15} and 4.2×10^{-16} over all 500 iterations; for R1 to 4.5×10^{-16} and 2.0×10^{-15}. All R1 and R2 runs (both seeds) reduce the functional to 25–34% of its first logged value (R2 consensus: ratio 0.250, final 6406.6; R1 consensus: ratio 0.316, final 8101.8), every final filter has unit norm, and the learned dictionaries give identical held-out test functionals (10 further colour images, 200 CBPDN iterations; R2 consensus 1524.7, R1 consensus 1880.2). The joint-sparsity representation R3, obtained by changing only the sparse-coding

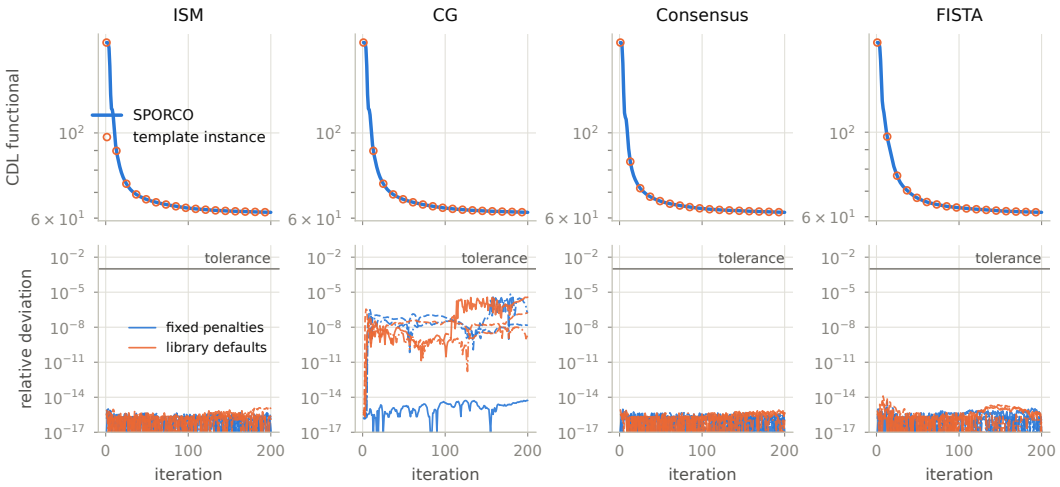


Figure 1: Template instances on SPORCO references (greyscale MIRFLICKR, $K = 5$, $M = 32$). Top: CDL functional of SPORCO (line) and of the template instance (circles), seed 0, fixed-penalty protocol. Bottom: relative deviation per iteration for three seeds (line styles) under the fixed-penalty protocol and SPORCO's library defaults; values below 10^{-17} are clipped.

Table 4: Equivalences derived from axis settings, verified on MIRFLICKR data ($K = 5$, 128×128, $M = 32$, 150 iterations). Deviation: maximum over the runs of each class of the maximum relative functional difference over all iterations, and of the relative Frobenius difference of the final dictionaries (for the contrast row: the *minimum* over its runs). Per-run values: Table 10.

	Identity (reference run ≡ template setting)	runs	Functional	Dictionary
E1	3D update (SPORCO, images as a volume) ≡ single blocks, image-DFT basis, σ/K	1	4.9×10^{-16}	1.4×10^{-14}
E2	R1 with channels as a volume (SPORCO, joint and consensus) ≡ channel-DFT cells, σ/C	2	8.7×10^{-16}	5.5×10^{-14}
E3	a basis acting inside blocks is inert (DFT, KLT, permutations; joint, channel, image, single)	8	4.4×10^{-16}	5.9×10^{-15}
E3	substitution ($\beta = 0$): FISTA for every partition and basis	4	3.0×10^{-16}	2.4×10^{-14}
E4	R2 (filter-joint norm) equivariant under orthonormal colour transforms (opponent, KLT)	4	7.5×10^{-16}	1.2×10^{-14}
E5	R1 colour CDL ≡ greyscale CDL on the CK channel images (ISM, consensus, FISTA)	3	0	0
contrasts	pairs where the axis algebra predicts *no* identity (minimum deviation)	8	1.5×10^{-2}	2.2×10^{-1}

stage, matches its SPORCO `DictLearn` reference to 1.6×10^{-15}. A second seed (other images and initial dictionary) is reported in Table 5; the variants with tuned penalties also match SPORCO run with the same penalties, to 1.3×10^{-13} in the functional and 1.1×10^{-12} in the dictionary (8 runs, both seeds).

Which representation fits colour better. At equal λ and M, R2 reaches a lower functional than R1 on the same images (6406.6 versus 8101.8 with consensus updates), and also a lower test functional. With correlated channels a single coefficient map per image and a 3-channel filter express a colour structure at ℓ_1 cost $\sqrt{3}$ times smaller than three equal per-channel maps; Sec. 6.5 shows that the gap closes and reverses as the channel correlation decreases.

6.4 Channel-aware splittings: validation and behaviour

Degenerate collapse. A new cell has no external reference, so we check it against the existing updates it must reduce to (Table 6): channel-coupled consensus with channel-group size one, on greyscale input, and on a single colour image; the channel-blocked partition on greyscale input and on

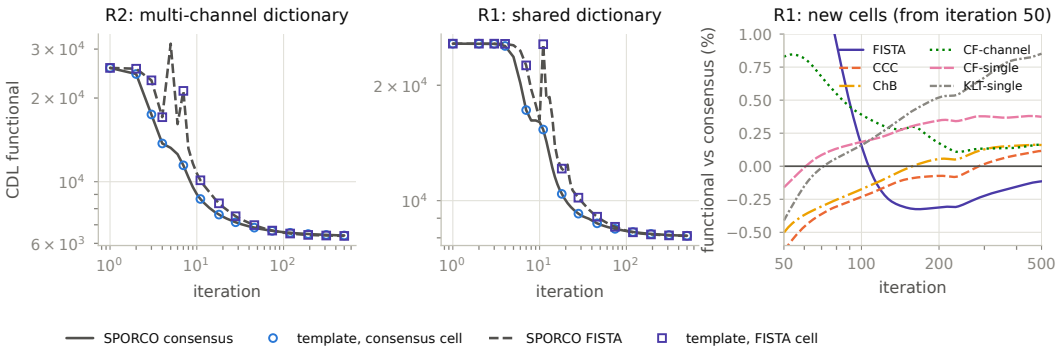


Figure 2: Colour CDL at the reference scale ($K = 40$, 256×256, $M = 64$, 500 iterations, seed 0). Left and middle: SPORCO references (lines) and template instances (markers) for the multi-channel-dictionary (R2) and shared-dictionary (R1) representations. Right: the new R1 cells and FISTA relative to the consensus cell (both with the reviewed paper's penalty rules), from iteration 50.

Table 5: Colour runs ($K = 40$, 256×256, $M = 64$, 500 iterations) for two seeds (different images and initial dictionaries). Deviation of the seed-1 template run from its seed-1 SPORCO reference (functional / dictionary; tuned variants against SPORCO run with the same tuned penalties), final training functional, and held-out test functional. The new cells have no SPORCO counterpart; their degenerate-collapse checks are in Table 6.

Instance	Deviation (seed 1)	F seed 0	F seed 1	test F seed 0	test F seed 1
R2 consensus	6.3×10^{-16} / 5.5×10^{-14}	6406.6	5676.1	1524.7	1820.1
R2 FISTA	2.5×10^{-15} / 1.5×10^{-13}	6383.1	5655.7	1528.0	1823.3
R1 consensus	4.1×10^{-16} / 2.5×10^{-14}	8101.8	7449.1	1880.2	2166.2
R1 FISTA	1.3×10^{-15} / 4.8×10^{-14}	8092.6	7436.3	1879.0	2163.3
R1 CCC (new)	no SPORCO counterpart	8111.2	7456.7	1879.7	2168.2
R1 ChB (new)	no SPORCO counterpart	8114.8	7466.5	1882.2	2171.0
R1 CF-channel (new)	no SPORCO counterpart	8115.2	7467.2	1882.1	2172.1
R2 consensus, tuned	1.8×10^{-15} / 6.6×10^{-14}	6412.6	5684.3	1527.1	1821.7
R2 FISTA, tuned	4.8×10^{-16} / 2.6×10^{-14}	6433.9	5655.3	1526.8	1825.3
R1 consensus, tuned	8.5×10^{-16} / 1.5×10^{-13}	8115.2	7449.9	1880.9	2165.8
R1 FISTA, tuned	4.9×10^{-14} / 4.4×10^{-14}	8344.7	7480.2	1892.4	2163.6

a single image; and the basis cells with $U = I$ and on greyscale input. All 9 checks reproduce actual SPORCO runs to 9.4×10^{-16} in the functional and 2.5×10^{-14} in the dictionary — round-off, not tolerance-level agreement — because at these settings the new cell *is* the old update in the iteration (5). The channel-Fourier cells are additionally reproduced by SPORCO in three-dimensional mode (E2, Table 4).

The dictionary update in isolation. To measure what the axes do to the subproblem they act on, we froze the coefficient maps of a colour CDL run (R1, 100 iterations) and solved (2) from the same random dictionary with every cell, each at the penalty that reaches a relative gap of 10^{-3} fastest (Fig. 3). On real RGB data every partition that merges observations converges faster than per-observation consensus: the channel-blocked and channel-Fourier blocked cells reach a gap of 10^{-4} in 51 and 57 iterations, against 153 for consensus, 88 for CCC and 76 for CCC-shuffled; the joint ISM update needs 31 and FISTA 40. CG with SPORCO's inner tolerance 10^{-3} stalls at a gap of 1.6×10^{-4}, the accuracy floor of its inexact solves. Sec. 7 explains the ordering of the consensus-type cells; the controlled-correlation panels show that CCC is as good as a random grouping of the same size when channels are independent and no better than consensus when they are strongly correlated.

Inside CDL. Interleaved with sparse coding, the advantage shrinks. At the reference colour scale (Fig. 2, right) the R1 cells end within 1.0% of each other after 500 iterations and their held-out test functionals within 0.5%. Early iterations differ more: at iteration 50, relative to the consensus cell (the paper's penalty), CCC is at -0.64%, the channel-blocked cell at -0.50%, and FISTA, which oscillates at first, at $+3.49\%$. With one inner iteration per stage the dictionary update is never solved to the accuracy at which the subproblem rates separate the cells, so the alternation, not the update,

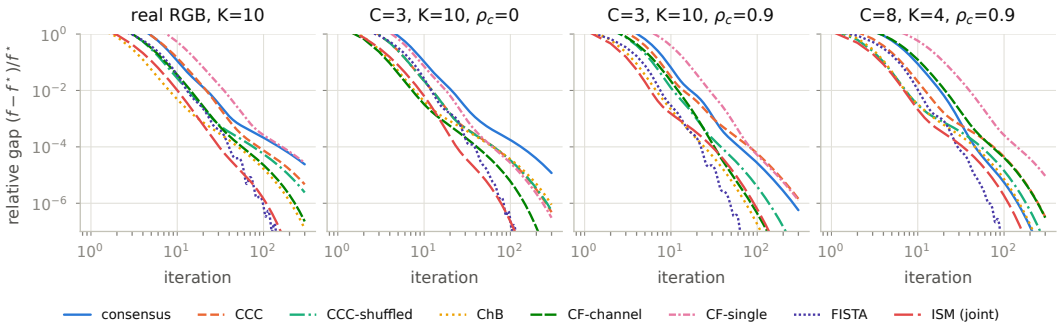


Figure 3: Dictionary-update subproblem with frozen coefficient maps: relative gap to the best value reached by any cell, each cell at its best penalty. Left: real colour MIRFLICKR ($K = 10$, 128×128, $M = 32$); others: controlled channel correlation.

Table 6: Internal consistency of the new cells: at a degenerate value of the new axis each cell must reproduce an actual SPORCO run (MIRFLICKR, 128×128, $M = 32$, 200 iterations, fixed-penalty protocol). P/r: number of blocks and block rank of the template instance. Deviations as in Table 3.

Cell	Degenerate setting	SPORCO counterpart	P/r	Functional	Dictionary	
CCC	channel-group size 1	R1 consensus	15/1	8.5×10^{-16}	7.5×10^{-15}	pass
CCC	$C = 1$ (greyscale)	consensus	5/1	9.4×10^{-16}	1.6×10^{-14}	pass
CCC	$K = 1$ (one colour image)	R1 ISM	1/3	6.3×10^{-16}	2.5×10^{-14}	pass
ChB	$C = 1$ (greyscale)	ISM	1/5	4.6×10^{-16}	2.1×10^{-14}	pass
ChB	$K = 1$ (one colour image)	R1 consensus	3/1	6.1×10^{-16}	1.3×10^{-14}	pass
KLT-single	$U = I$	R1 consensus	15/1	8.5×10^{-16}	6.8×10^{-15}	pass
KLT-single	$C = 1$	consensus	5/1	9.4×10^{-16}	1.6×10^{-14}	pass
CF-single	$U = I$	R1 consensus	15/1	8.7×10^{-16}	6.1×10^{-15}	pass
CF-single	$C = 1$	consensus	5/1	5.6×10^{-16}	1.5×10^{-14}	pass

sets the pace. One cell falls outside this band: FISTA with the step chosen by the grid search at $K \leq 20$ ($L = 420$) is too aggressive at $K = 40$, oscillates, and ends $+3.0\%$ above consensus, while the reviewed paper's rule $L = 14CK$ is stable; the tuned consensus penalty likewise ends $+0.17\%$ from the paper's rule. Rules fitted on short runs at small K do not transfer reliably. The exact-without-multiplier probe, which Proposition 2(iii) predicts to be biased, stalls at a relative gap of 1.1×10^{-2} on the subproblem.

6.5 Which representation and which update, as a function of C, K and channel correlation

We swept the number of channels and images, $(C, K) \in \{1, 3\} \times \{4, 16, 32\}$ and $\{8\} \times \{4, 16\}$, and the channel correlation. Controlled data mix independent high-passed MIRFLICKR greyscale images with the Cholesky factor of an equicorrelation matrix with parameter $\rho_c \in \{0, 0.5, 0.9\}$ ($\{0, 0.9\}$ for $C = 8$); because the source images differ in energy, the measured mean inter-channel correlation is lower than ρ_c (0.33–0.52 for $\rho_c = 0.5$, 0.76–0.91 for $\rho_c = 0.9$). Real RGB (0.84–0.88) is added for $C = 3$. Images are 128×128, $M = 32$, 200 iterations. Every cell was first run at three penalties ($\times\frac{1}{3}$, $\times1$, $\times3$ the tuned rule); in 101 of the 236 (cell, regime) pairs the best value lay on the edge of this grid, so the grid was extended in that direction by further factors of 3 until the final functional stopped improving (105 extra runs, after which no best value lies on an edge). The joint ISM cell was first run only for $CK \leq 48$ to bound its runtime and was then added, with the same grid procedure, for the six regimes with $CK > 48$. For each regime and representation we record the best final functional $F^\star$ over all cells, the iterations each cell needs to reach $(1 + \epsilon)F^\star$, and the corresponding time from per-iteration costs measured on an otherwise idle GPU (Table 8). Fig. 4 and Table 7 summarise the result for $\epsilon = 1\%$.

The channel correlation picks the representation. The sign of $F^\star_{\mathrm{R2}} - F^\star_{\mathrm{R1}}$ follows the channel correlation in every configuration (Fig. 4, right). With independent channels the shared dictionary fits better, by 6.9–24.1%, and more so with $C = 8$: per-channel maps place structure independently in each channel, whereas the single map of R2 ties the locations and amplitudes of all channel slices

Table 7: Multi-channel regimes of the regime study (128×128, $M = 32$, 200 iterations, seed 0). Channels: controlled mixing parameter ρ_c or real RGB; corr.: measured mean inter-channel correlation of the training signals. $F^\star$: best final functional over all cells of the representation (cell achieving it). Gap: $(F^\star_{\mathrm{R2}} - F^\star_{\mathrm{R1}})/F^\star_{\mathrm{R1}}$. Fastest: cell that first reaches $1.01\,F^\star$ of its representation, and its time (s) from per-iteration costs measured on an idle GPU.

C	K	channels	corr.	$F^\star_{\mathrm{R1}}$		$F^\star_{\mathrm{R2}}$		gap (%)	fastest R1	fastest R2
3	4	ρ_c=0.0	0.00	218.6	ISM	233.7	ISM	+6.9	FISTA (0.43)	ISM (0.40)
3	4	ρ_c=0.5	0.33	212.7	FISTA	208.8	FISTA	−1.8	FISTA (0.39)	FISTA (0.36)
3	4	ρ_c=0.9	0.76	155.2	CCC-shuffled	131.0	ISM	−15.6	CCC-shuffled (0.37)	FISTA (0.38)
3	4	RGB	0.88	119.7	ChB	91.0	FISTA	−24.0	FISTA (0.39)	FISTA (0.43)
3	16	ρ_c=0.0	0.00	954.9	ISM	1047.9	FISTA	+9.7	FISTA (0.91)	FISTA (0.55)
3	16	ρ_c=0.5	0.52	1067.0	FISTA	1002.9	ISM	−6.0	FISTA (0.83)	FISTA (0.60)
3	16	ρ_c=0.9	0.91	1042.3	FISTA	793.6	FISTA	−23.9	FISTA (0.86)	FISTA (0.58)
3	16	RGB	0.84	1009.9	FISTA	809.0	ISM	−19.9	FISTA (0.84)	FISTA (0.59)
3	32	ρ_c=0.0	0.00	1943.8	FISTA	2140.1	ISM	+10.1	FISTA (1.63)	FISTA (0.81)
3	32	ρ_c=0.5	0.51	2149.4	FISTA	2033.9	ISM	−5.4	FISTA (1.50)	FISTA (0.84)
3	32	ρ_c=0.9	0.91	2112.5	FISTA	1611.3	ISM	−23.7	FISTA (1.53)	FISTA (0.84)
3	32	RGB	0.87	1931.5	FISTA	1532.0	FISTA	−20.7	FISTA (1.84)	FISTA (0.98)
8	4	ρ_c=0.0	0.00	655.8	FISTA	792.9	ISM	+20.9	FISTA (0.67)	ISM (0.93)
8	4	ρ_c=0.9	0.78	460.3	ChB	330.3	ISM	−28.2	ChB (0.64)	ISM (0.96)
8	16	ρ_c=0.0	0.00	2545.0	FISTA	3158.3	ISM	+24.1	FISTA (2.05)	FISTA (1.16)
8	16	ρ_c=0.9	0.91	2884.1	FISTA	1745.7	FISTA	−39.5	FISTA (2.01)	FISTA (1.27)

of an atom. Already at $\rho_c = 0.5$ the multi-channel dictionary is ahead, by 1.8–6.0%; with strongly correlated channels it is ahead by 15.6–39.5%, and on real RGB by 19.9–24.0%, in line with the reference-scale colour runs (Sec. 6.3). For $C = 3$, $K = 4$ the crossover lies between measured correlations 0 and 0.33 (+6.9% and −1.8%).

Iteration counts are similar; cost picks the update. Within a regime, the cells (each at its best penalty) need similar numbers of iterations to come within 1% of $F^\star$: between 90 and 193 over all multi-channel R1 regimes, and in the median regime the slowest cell needs $1.39\times$ the iterations of the quickest. Their final functionals differ by at most 0.9% (median 0.4%). The ranking in time is therefore set mostly by the cost per iteration. FISTA, whose update needs no linear solve, reaches the target first in all 3 greyscale regimes ($K = 4, 16, 32$; consensus takes 1.02–$1.25\times$ its time, CG 1.48–$1.90\times$ and ISM 1.17–$5.78\times$), in 14 of the 16 multi-channel R1 regimes (and within $1.13\times$ of the fastest cell in the other two), and in 13 of the 16 R2 regimes; the R1 count is 14 and 11 for $\epsilon = 0.5\%$ and 2%. The exact joint updates are never fastest in R1: ISM, whose rank-CK systems are the most expensive, takes $9.17\times$ the fastest time in the median regime (up to $23.3\times$), and CG $1.49\times$. In R2, whose dictionary-update rank stays K (Proposition 5), ISM is the fastest cell in 3 of the 6 regimes with $K = 4$ and in none with $K \geq 16$.

The channel-aware splittings follow the cost model and the homogeneity rule. The multiplier-coupled R1 cells with small blocks are close to each other: in the median regime consensus, CCC and CCC-shuffled take $1.27\times$, $1.29\times$ and $1.21\times$ the fastest time, and CCC-shuffled is the fastest cell for $C = 3$, $K = 4$, $\rho_c = 0.9$. The rank-K channel blocks are competitive with few images and fall behind as K grows, as their cost predicts: ChB takes 1.0–$1.2\times$ the fastest time at $K = 4$ (it is the fastest cell for $C = 8$, $K = 4$, $\rho_c = 0.9$), 1.9–$2.3\times$ at $K = 16$ and 3.0–$3.5\times$ at $K = 32$. The channel-basis cells behave as the homogeneity rule of Sec. 7 predicts: CF-single takes $1.40\times$ the fastest time with independent channels but $2.01\times$ with correlated or RGB channels, where the channel DFT concentrates the energy in one component and leaves weak blocks (KLT-single: $1.42\times$ and $2.03\times$).

Selection map. Choose the representation by the channel correlation — the multi-channel dictionary for RGB and other correlated channels, the shared dictionary for nearly independent ones. With one inner iteration per stage, choose the cheapest consistent update: FISTA, or, where an update that parallelises over images is wanted, consensus or CCC. Exact joint solves pay off only for R2 with few images, and channel blocks only with few images; the subproblem advantage of the channel-blocked splittings (Sec. 6.4) does not survive the alternation at this coupling.

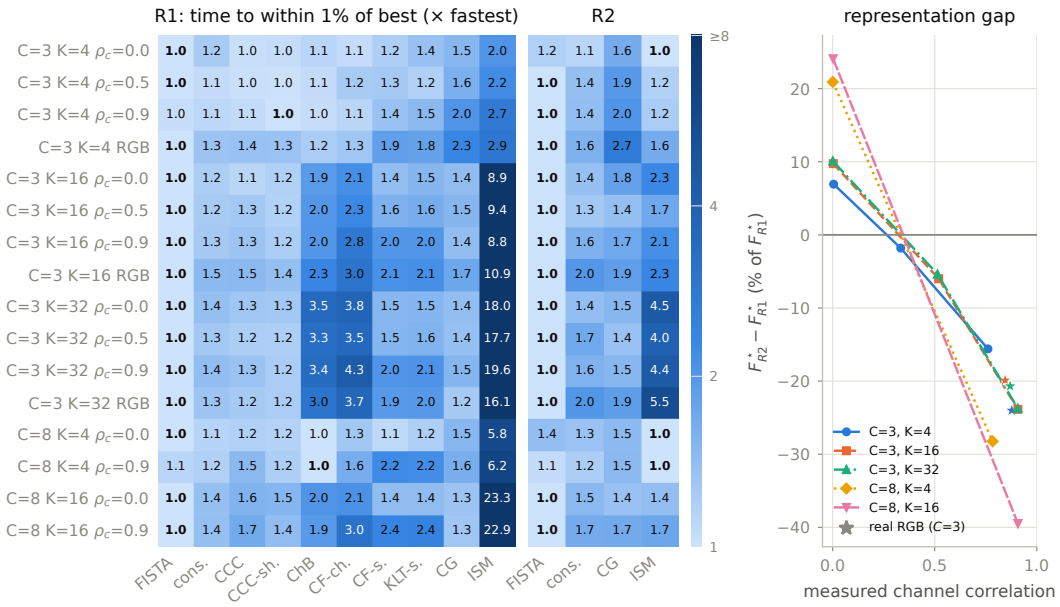


Figure 4: Selection map. Left and middle: time for each cell to come within 1% of the best final functional of its representation in each regime, relative to the fastest cell of that regime (bold: fastest; shading: $\log_2$ of the ratio). Rows: C, K and the controlled mixing parameter ρ_c or real RGB. Right: functional gap between the representations, $(F^\star_{\mathrm{R2}} - F^\star_{\mathrm{R1}})/F^\star_{\mathrm{R1}}$, against the measured channel correlation (lines: controlled data, $\rho_c = 0, 0.5, 0.9$; stars: real RGB).

7 Mechanism: what the partition and the basis change

With the exact local model and a multiplier, the cells of one row of Table 2 solve the same local systems up to grouping, so their differences are differences in how fast the copies reach consensus. We study this at three levels of modelling, with each cell at its best penalty (Fig. 5).

A frequency-decoupled model points the wrong way. Drop the whole constraint and look at one frequency: each observation contributes a rank-one curvature $\overline{\mathbf{a}_j}\mathbf{a}_j^\top$, and the multiplier iteration has linear error dynamics on $(g, h_1, \ldots, h_P)$ whose spectral radius is the convergence factor. In this model, copies that observe the *same* direction must be reconciled through the multiplier: if P blocks see one direction with curvature q, the disagreement modes decay with factor $q/(q+\sigma)$ while the mean decays with $\sigma/(q+\sigma)$, so the factor is at least $\frac{1}{2}$, whereas merging them gives $\sigma/(Pq+\sigma) \to 0$. Correlated colour channels are nearly collinear rows, so the model predicts that grouping an image's channels (CCC) is the right partition. On coefficient spectra of a real colour CDL run (panel a; 32 frequencies, one penalty per cell shared by all frequencies) it indeed ranks CCC (58 iterations per decade at the median frequency) far ahead of consensus (253) and of the same block sizes without alignment (CCC-shuffled, 114); on synthetic rows the gap between CCC and CCC-shuffled opens as the channel correlation grows (Appendix, Fig. 6).

The support constraint couples the frequencies. The filters live on an 8×8 support, and the projection onto it couples all frequencies; on that support a single observation's curvature is no longer rank one but spans every filter the observation uses. Keeping the support constraint and dropping only the unit norm gives an affine iteration whose rate we measure directly (panel b; $K = 6$ real colour images, 64×64, $M = 16$): consensus, CCC and CCC-shuffled now converge at the same rate (137, 140 and 140 iterations per decade), the optimal penalty of CCC being that of consensus scaled by the block size ($\sigma = 1 \to 3$), while the channel-blocked partition is about twice as fast (71). Merging redundant observations is then only a penalty rescaling; what changes the rate is how *different* the blocks are.

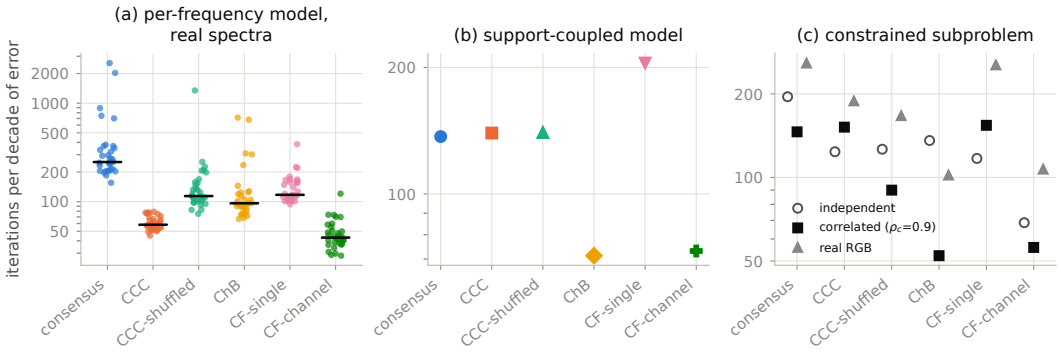


Figure 5: Convergence of consensus-type cells (iterations per decade of error, lower is faster; each cell at its best penalty). (a) Frequency-decoupled linear model on coefficient spectra of a real colour CDL run (32 frequencies, one penalty per cell shared across frequencies; bars: medians). (b) Support-coupled linear model (support constraint kept, unit norm dropped). (c) Constrained dictionary-update subproblem with frozen maps: asymptotic rate for independent, correlated and real RGB channels ($K = 10$, 128×128, $M = 32$).

Block homogeneity governs the constrained subproblem. Panel (c) reports the asymptotic rate of the actual dictionary-update subproblem (frozen maps, full constraint, last 100 of 300 iterations) in three regimes. With independent channels, CCC and CCC-shuffled coincide (124 and 126) and beat consensus (195): grouping C unrelated channel images makes each block more representative of the data. With strongly correlated channels the picture flips: CCC groups near-copies and gains nothing over consensus (152 versus 146), whereas grouping channels of *different* images does (90), and the channel-blocked partition, whose C blocks each contain every image, is the fastest consensus-type cell (52). On real RGB data the channel-blocked cells again lead and CCC-shuffled again beats CCC: channel-blocked 102, channel-Fourier blocked 107, CCC-shuffled 167, CCC 189, consensus 258 (ISM 53 and FISTA 65 for reference). With eight correlated channels but only four images (Fig. 3, right), the channel-blocked partition merely ties consensus (76 versus 77) and CCC is slower (103): with few images no block can be representative. Bases act through the same rule: the channel DFT turns independent channels into mixtures, each representative of the image, and CF-single beats consensus (117 versus 195); with correlated channels it concentrates the energy in one component and leaves weak, unrepresentative blocks, and CF-single is no better than consensus (254 on RGB; 205 versus 137 in the support-coupled model). The rule that emerges is the one familiar from federated and decentralised optimisation [24, 20]: consensus is slow when the copies' data disagree, so a good partition gives every block a representative sample — split correlated channels *by channel* (each block sees all images), and mix independent channels (channel DFT) before splitting. The frequency-decoupled intuition that redundant observations should share a block is an artefact of ignoring the support constraint.

8 Where the abstraction is incomplete

Updates that are not cells. *Spatial tiling* arranges the K images into one large image so that the update has rank one. This is not a unitary change of basis on the observation index: filters that straddle a tile boundary read the neighbouring tile instead of wrapping around their own image, so tiling solves a slightly different data term. The iteration can run that data term, but the result is not the dictionary update (2), so we do not count tiling as recovered. *Masked* dictionary updates, $\frac{1}{2}\|W(X\mathbf{d} - \mathbf{s})\|^2$, lose the per-frequency structure (3). The isotropic cells carry the mask without change of axes (the gradient maps the residual to the spatial domain, weights it by W^2 and maps it back), and the template's masked FISTA cell reproduces SPORCO's masked dictionary learning to 8.0×10^{-16} (three seeds, 25% of the pixels masked; Table 3). The exact cells, however, would need the block-constraint or extended-consensus splitting of [12], i.e. a second lifting (of the residual, as in mask decoupling [1]); that lifting is not one of our axes, so the template refuses exact cells with a mask (attempts log).

Cells that are not consistent or not closed-form. Proposition 2 separates the grid. With substitution ($\beta = 0$) the projection must be taken in the metric of the local model; it has a closed form only for an isotropic model, because C_{PN} is a support-and-norm set in the *spatial* domain whereas the exact curvature is diagonal in the *frequency* domain. This is why FISTA uses a scalar step, why frequency-diagonal majorizers cannot be combined with exact projection, and why the substitution row of the grid collapses to one method. The remaining corner — exact local model without a multiplier — runs, but converges to fixed points of a Moreau-smoothed problem; Sec. 6.4 shows the resulting bias.

Recovery that is not bit-faithful. CG agrees with the reference only to 6.8×10^{-6} (functional) and 4.4×10^{-4} (dictionary). The CG stopping test $\|r\| < \tau\|b\|$ is discontinuous in round-off, so an outer iteration may run one CG step more or fewer than the reference, and the alternation propagates the difference. Tightening both tolerances to 10^{-6} reduces the disagreement to 7.3×10^{-9} / 6.5×10^{-7}, consistent with this explanation.

Conventions that must be mapped. Equivalences hold only with matched conventions: the 3D penalty maps as $\sigma \mapsto \sigma/K$ (or σ/C), the multiplier initialisation must be mapped with it, and the reference classes initialise their multipliers differently ($H_0 = G_0$ versus $H_0 = G_0/\sigma$). For complex bases our implementation supports fixed penalties only (residual balancing would need the half-spectrum norms of complex copies), and for R2 the reference implementation normalises filters jointly over channels, which is not the per-channel constraint written in Eq. (80) of the reviewed paper; the per-channel version is available as an axis value but is not the one matched against SPORCO.

What the template does not cover. The template abstracts the dictionary update only. Proposition 5(b) shows that the same cells are sparse-coding solvers for the multi-channel-dictionary representation (e.g. consensus over channels for large C), but we hold the sparse-coding stage fixed and do not run them. The coupling between the two stages (one inner iteration each, auxiliary variables exchanged) is also held fixed; several of our conclusions about the size of subproblem gains are specific to that coupling.

9 What an iteration costs

Per-iteration cost from the block structure. Per outer iteration and dictionary channel, every cell spends $C_x KM$ FFTs on the coefficient spectra and two batches of M FFTs on the consensus variable; the cells differ in the local solves. With P blocks of rank r, the iterated Sherman–Morrison recursion costs $O(r^2)$ inner products of length M per block and frequency, i.e. $O(|J|\, r\, MN)$ in total, and its memory is $O(PMN)$ for copies and multipliers plus $O(rMN)$ for the recursion. For R1 ($|J| = CK$) the joint partition therefore costs $O(C^2K^2MN)$, CCC $O(C^2KMN)$, the channel-blocked partition $O(CK^2MN)$, and consensus or FISTA $O(CKMN)$; for R2 the joint factor is shared by the C channels and costs $O(K^2MN)$, while sparse coding costs $O(C^2KMN)$ because its systems have rank C (Proposition 5). Table 8 reports measured per-iteration times of both stages.

Measured costs. Table 8 confirms the block-rank model. The dictionary updates of consensus and CCC cost about the same (3.7 and 3.7 ms at $C = 3$, $K = 16$), FISTA is the cheapest (2.0 ms; no linear solve), the rank-K channel-blocked update grows quadratically in K (8.7 ms at $K = 16$, 33.1 ms at $K = 32$ against 7.2 ms for consensus), and the joint ISM update is the most expensive (65.3, 246.7 and 431.2 ms for 48, 96 and 128 observations). The channel-blocked update costs $1.8\times$ consensus and $0.16\times$ the joint ISM update at $K = 10$, where it needs $2.5\times$ fewer iterations than consensus on the subproblem (Sec. 7), but $4.6\times$ consensus at $K = 32$. At the reference colour scale ($K = 40$, 256×256, $M = 64$) an R1 iteration spends 127.9 ms in sparse coding and 69.1 ms in the consensus update (34.4 ms for FISTA, 65.4 ms for CCC, 238.5 ms for the channel-blocked update), and an R2 iteration 51.3 ms and 58.3 ms. In the sparse-coding stage R2 is cheaper than R1 at every tested C (3.1 versus 6.2 ms at $C = 3$, 5.5 versus 16.9 ms at $C = 8$, $K = 16$), because R1 transforms C times more coefficient maps; R2's rank-C recursion, which the $K \leftrightarrow C$ duality identifies as the place where its channel count enters, is not yet the dominant term at $C \leq 8$.

Budget used. The run times recorded in the logs add up to 3.9 GPU hours (colour runs, regime study, subproblem study, penalty search; one GPU) and 1.2 CPU hours for the SPORCO comparisons

Table 8: Measured time per outer iteration (ms) on one GPU, otherwise idle: dictionary update of each cell (R1 and R2) and the sparse-coding stage (CBPDN) of each representation; 128×128 images, $M = 32$, mean of 20 iterations after 5 warm-up iterations (joint ISM for $CK > 48$ measured in a separate run on the same idle GPU).

Stage	C=3, K=4	C=3, K=16	C=3, K=32	C=8, K=4	C=8, K=16
R1 (shared dictionary)					
sparse coding	2.0	6.2	13.1	4.3	16.9
D-update, FISTA	1.2	2.0	3.4	1.5	4.3
D-update, consensus	1.3	3.7	7.2	2.6	9.4
D-update, CCC	1.5	3.7	6.7	3.5	11.9
D-update, ChB	1.7	8.7	33.1	2.7	17.8
D-update, CF-channel	1.9	9.3	34.3	3.4	20.6
D-update, CF-single	1.8	5.5	10.8	3.2	11.7
D-update, CG	3.5	5.2	8.0	4.3	9.6
D-update, ISM	5.5	65.3	246.7	29.0	431.2
R2 (multi-channel dictionary)					
sparse coding	1.9	3.1	5.4	4.2	5.5
D-update, FISTA	1.7	2.1	2.4	4.1	4.4
D-update, consensus	1.6	3.3	5.6	4.7	8.6
D-update, CG	4.8	6.7	7.3	8.6	11.2
D-update, ISM	2.2	9.3	32.3	4.6	11.9

on the CPU (recovery and degenerate-collapse runs, at most 16 threads), plus the equivalence and mechanism studies, whose per-run times were not logged.

10 Related work

Dictionary updates for CDL. Frequency-domain ADMM for convolutional sparse coding and dictionary learning originates with Bristow et al. [4]; Wohlberg introduced the Sherman–Morrison solution for sparse coding and the ISM and CG inner solvers for the dictionary update [29, 31]. Šorel and Šroubek derived consensus and "3D" updates via the matrix inversion lemma [27]; Heide et al. proposed a splitting with a spatial mask [14]; Garcia-Cardona and Wohlberg compared these updates, introduced FISTA and mask-decoupled hybrids, related 3D to consensus, and studied penalty scaling [12], building on their analysis of subproblem coupling [11]. Chun and Fessler cast CDL as block proximal gradient with majorizers (BPG-M), with diagonal majorizer designs and momentum [6]; our isotropic local model is the scalar case of their majorizer axis, and the requirement that the majorizer be isotropic whenever the coupling is by substitution (Sec. 8) explains why their dictionary majorizers act in the spatial domain. Silva and Rodríguez combined accelerated proximal gradient with consensus for the dictionary update [25]; in our grid this is the multiplier-coupled cell with an isotropic local model. Choudhury et al. partition large training sets into blocks with local dictionaries tied by consensus [5] — the image-group partition. Local-processing [22], online [18, 19, 28], and distributed [21] CDL change the coupling or the data access pattern and are outside the batch template studied here. SPORCO [33] implements the batch updates we recover.

Multi-channel convolutional representations. Wohlberg described the single-channel-dictionary, multi-channel-dictionary and joint-sparsity representations of colour images [30]. The reviewed comparison derives the multi-channel-dictionary update, notes that it costs no more than the greyscale one, and identifies the $K \leftrightarrow C$ duality [12]. Garcia-Cardona and Wohlberg's subsequent multi-channel study extends the dictionary updates and sparse coding to signals with many channels, introduces a product of a convolutional and a standard dictionary for hyperspectral data, and evaluates the methods on hyperspectral imagery [13]. Our work does not propose a new representation and does not re-benchmark those extensions: it supplies the iteration of which the updates are instances, derives their equivalences, shows that the tying map decides on which side of the alternation the channel rank falls, and fills cells of the resulting grid (channel-coupled, channel-blocked and channel-basis splittings) that the earlier work does not contain. The product dictionary of [13] with an orthogonal channel factor is R1 in a rotated colour basis; our KLT cells instead keep the R1 problem fixed and rotate only the splitting. Multi-modal online CDL [10] and hyperspectral convolutional models with three-dimensional filters [5] use other representations.

Operator splitting and majorization. The iteration (5) combines standard ingredients: global-variable consensus ADMM [3], over-relaxation and residual balancing [3, 32], linearised ADMM [34], majorization–minimisation [16, 26], and accelerated proximal gradient [2]. Unifications of proximal algorithms as instances of a few splittings are classical [7, 23, 8, 9]; our contribution is the specific lift for convolutional dictionary updates in which the basis, the partition and the tying map are explicit axes, so that the published updates, their equivalences and new channel-aware splittings are read off one formula and executed by one implementation. Convergence of consensus ADMM in decentralised settings depends on how the data are spread across agents [24, 20], which is the mechanism behind the partition and basis effects in Sec. 7.

11 Limitations and conclusion

Limitations. The template covers the dictionary update with the sparse-coding stage and the one-iteration coupling held fixed; masked data are supported only by the isotropic local model, tiling is not instantiated, and the dual sparse-coding cells predicted by Proposition 5(b) are not run. Our convergence analysis consists of linear models; the support-coupled one reproduces the ordering of the constrained subproblem, but it drops the unit-norm constraint and is not a convergence proof for the non-convex alternation. The penalty rules for the new cells come from grid searches on one data source at two training-set sizes (refined per regime in the regime study), and such rules do not transfer reliably to larger K (Sec. 6.4); all comparisons use one machine and one GPU implementation, so absolute times are implementation-specific even though the cells share one code path. The colour study uses RGB images; the controlled-correlation data approximate, but are not, hyperspectral images.

Conclusion. Batch convolutional dictionary updates are one iteration with six axes. Writing the iteration down turns comparisons into statements about axis values: which updates coincide (3D and image-DFT consensus, channel volumes and channel-DFT consensus, bases inside blocks, partitions under substitution), which corners are inconsistent (exact local models without a multiplier), and on which side of the alternation a representation puts the channel rank (the dictionary update for a shared dictionary, sparse coding for channel dictionaries). The same structure predicts channel-aware splittings that are absent from existing implementations. They are exact where they must be (degenerate collapse onto SPORCO), and on the dictionary-update subproblem the channel-blocked splittings converge $2.5\times$ faster than consensus at $0.16\times$ the per-iteration cost of the joint update; the reason is block homogeneity, not the within-block redundancy that a frequency-decoupled analysis suggests. Inside CDL with one inner iteration per stage these subproblem gains mostly wash out: across 16 multi-channel regimes the updates need similar iteration counts, so the cost per iteration (set by block ranks and the $K \leftrightarrow C$ duality) selects the update, and the channel correlation selects the representation.

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A Reproducibility details

Code and logs. `proposal/code/mci/` contains the template (`template.py`), the inner solvers (`solvers.py`), the cell registry (`cells.py`), the CDL loop and SPORCO reference runner (`cdl.py`), the GPU construction of SPORCO's dictionary-update modules (`sporco_gpu.py`) and the MIR-FLICKR pipeline (`data.py`). Every experiment script in `proposal/code/experiments/` writes JSON (and NPZ for dictionaries) to `proposal/logs/`; all numbers in this paper are produced from those logs by `make_tables.py` and all figures by the `fig_*.py` scripts. The template, its axes, the planned recovery targets, representations and new cells were fixed in `framework_plan.md` before any validation run; deviations are listed in `attempts_log.md`.

Reference configuration. SPORCO 0.2.3.dev0 from the provided source tree, NumPy 2.5, SciPy 1.16, pyFFTW 0.15 (16 threads at most), CuPy 14.2 on one NVIDIA H200 GPU (as recorded in the environment block of the GPU run logs). Recovery runs are float64 on the CPU; colour runs are float64 on the GPU. The CG settings are SPORCO's (`StopTol` 10^{-3}, `MaxIter` 1000, warm start). FISTA uses Nesterov momentum without backtracking. SPORCO's effective options (relaxation 1.8, residual balancing enabled for ISM/CG and disabled for consensus by its option inheritance) are read from SPORCO's option objects and logged with every reference run.

Penalty rules. Table 9 lists the penalties used; the rules for the new cells and the "tuned" variants are the output of `select_params.py` on the grid searches in `logs/tuning/`.

Table 9: Penalty rules used (CBPDN penalty $\rho = 2.2$ for joint and FISTA cells, 3.0 otherwise). "Paper" rules are the reviewed paper's guidelines (with its colour adjustment $L = 3 \times 14K$ for R2); "tuned" rules come from the grid searches at $K \in \{10, 20\}$ ($n =$ number of observations: CK for R1, K for R2). The new cells have no published rule; for R2 ISM the search selected the paper rule.

Cell	paper rule	tuned rule
R1 consensus	$\sigma = 2.2$	$\sigma = 0.707$
R1 ISM	$\sigma = 0.5n + 7$	$\sigma = 0.783\,n$
R1 FISTA	$L = 14n$	$L = 3.5\,n$
R1 CCC (new)	none published	$\sigma = 2.2$
R1 ChB (new)	none published	$\sigma = 0.5\,K$
R1 CF-single (new)	none published	$\sigma = 0.707$
R1 CF-channel (new)	none published	$\sigma = 1.05\,K$
R1 KLT-single (new)	none published	$\sigma = 0.354$
R1 KLT-channel (new)	none published	$\sigma = 1.05\,K$
R2 consensus	$\sigma = 2.2$	$\sigma = 1.48$
R2 ISM	$\sigma = 0.5K + 7$	same as paper rule
R2 FISTA	$L = 42K$	$L = 21\,K$

B All equivalence runs

Table 10: Every equivalence run of Table 4. Rows marked † are contrasts where the axis algebra predicts *no* identity. E4 rows compare a run in RGB with the same run in a rotated colour space (dictionary mapped back before comparison).

	Reference run	Template / second run	Functional	Dictionary
E1	3D (SPORCO, volume over images)	single, DFT_{img}, σ/K	4.9×10^{-16}	1.4×10^{-14}
E1†	3D (SPORCO)	single, identity (consensus)	3.8×10^{-2}	4.9×10^{-1}
E2	R1 volume over channels, joint (SPORCO ISM)	channel, DFT_{ch}, σ/C	6.4×10^{-16}	8.1×10^{-15}
E2†	R1 volume over channels, joint (SPORCO ISM)	channel, identity (ChB)	2.1×10^{-2}	2.2×10^{-1}
E2	R1 volume over channels, consensus (SPORCO)	single, DFT_{ch}, σ/C	8.7×10^{-16}	5.5×10^{-14}
E2†	R1 volume over channels, consensus (SPORCO)	single, identity (consensus)	1.5×10^{-2}	3.7×10^{-1}
E3	CCC (image, identity)	image, DFT_{ch}	3.0×10^{-16}	5.3×10^{-15}
E3	CCC (image, identity)	image, KLT_{ch}	4.4×10^{-16}	3.6×10^{-15}
E3	ISM (joint, identity)	joint, DFT_{ch}	3.0×10^{-16}	4.3×10^{-15}
E3	ISM (joint, identity)	joint, KLT_{ch}	3.0×10^{-16}	4.3×10^{-15}
E3	ISM (joint, identity)	joint, DFT_{img}	3.2×10^{-16}	4.6×10^{-15}
E3	FISTA (β=0)	single, identity	3.0×10^{-16}	1.3×10^{-14}
E3	FISTA (β=0)	image, identity	3.0×10^{-16}	2.0×10^{-14}
E3	FISTA (β=0)	channel, DFT_{ch}	3.0×10^{-16}	2.0×10^{-14}
E3	FISTA (β=0)	single, KLT_{ch}	3.0×10^{-16}	2.4×10^{-14}
E3	ISM (joint, identity)	joint, images permuted per channel	3.0×10^{-16}	5.9×10^{-15}
E3	ChB (channel, identity)	channel, images permuted per channel	3.0×10^{-16}	3.1×10^{-15}
E3	consensus (single, identity)	single, images permuted per channel	3.0×10^{-16}	1.7×10^{-15}
E3†	CCC (image, identity)	image, images permuted per channel	1.6×10^{-2}	4.9×10^{-1}
E4	R2 consensus (SPORCO), RGB	same, opponent colour space	5.6×10^{-16}	1.1×10^{-14}
E4	R2 FISTA (SPORCO), RGB	same, opponent colour space	7.4×10^{-16}	1.2×10^{-14}
E4†	R2, per-channel norm, RGB	same, opponent colour space	2.7×10^{-1}	1.1×10^{0}
E4†	R1 consensus (SPORCO), RGB	same, opponent colour space	2.6×10^{-1}	3.9×10^{-1}
E4	R2 consensus (SPORCO), RGB	same, KLT colour space	7.5×10^{-16}	7.0×10^{-15}
E4	R2 FISTA (SPORCO), RGB	same, KLT colour space	7.3×10^{-16}	1.2×10^{-14}
E4†	R2, per-channel norm, RGB	same, KLT colour space	2.7×10^{-1}	1.1×10^{0}
E4†	R1 consensus (SPORCO), RGB	same, KLT colour space	2.6×10^{-1}	3.8×10^{-1}
E5	R1 ISM (SPORCO), colour	greyscale ISM on CK images	0	0
E5	R1 consensus (SPORCO), colour	greyscale consensus on CK images	0	0
E5	R1 FISTA (SPORCO), colour	greyscale FISTA on CK images	0	0

C Frequency-decoupled model on synthetic rows

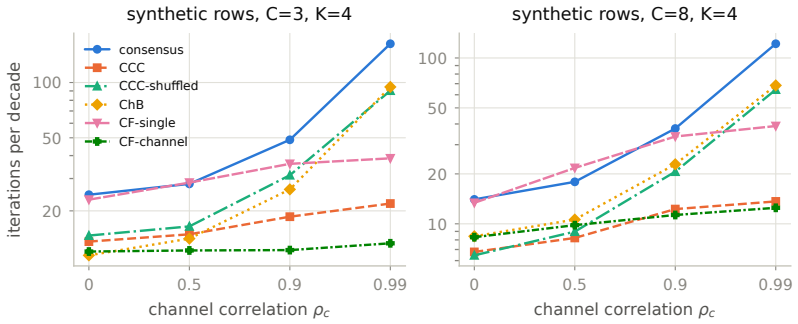


Figure 6: Frequency-decoupled linear model (one frequency, no constraint) on synthetic rows $\mathbf{a}_{c,k} = \sqrt{\rho_c}\,\mathbf{u}_k + \sqrt{1-\rho_c}\,\mathbf{v}_{c,k}$ ($M = 8$, $K = 4$; mean of three draws, optimal penalty per configuration). In this model aligned grouping (CCC) separates from a random grouping of the same size (CCC-shuffled) as the channel correlation grows; Sec. 7 shows that the filter-support constraint removes this effect.